Disabling a tool

MCC generates the G code of a drawing **for every tool in the drawing**; it does not output only the selected tool. To skip the operations of one tool temporarily you do not need to delete or regenerate the G code: **disable the tool**, and the machine skips that tool's section at run time.

How it looks in the G code

In the generated program every tool starts its own block. The first line of the block is a jump condition: if the tool's disable parameter is **not 0**, the NC skips all lines of that tool and continues with the next tool's block.

```
N01000 @121 R154 K0 K2000     ; if R154 ≠ 0 jump to N02000 (tool 4)
...                           ; tool 4 operations
N02000 @121 R156 K0 K3000     ; if R156 ≠ 0 jump to N03000 (tool 6)
...                           ; tool 6 operations
N03000 M17
```

On B&R controlled machines the same logic is written as `IF R... <> 0` / `GOTO N...000` / `ENDIF`.

Parameter number = 150 + tool number.

Tool	Parameter		Tool	Parameter
T1	R151		T11	R161
T2	R152		T12	R162
T3	R153		T13	R163
T4	R154		T14	R164
T5	R155		T15	R165
T6	R156		T16	R166
T7	R157		T17	R167
T8	R158		T18	R168
T9	R159		T19	R169
T10	R160		T20	R170

Method 1 — Tool Disable box

On machines whose tool offset screen has a **Tool Disable** box (e.g. CN779, CN784, DCM019B):

1. Open the tool offset screen and select the tool.
2. Tick **Tool Disable** and save.
3. To use the tool again, clear the box and save.

Method 2 — R parameter

On machines without the Tool Disable box (e.g. CN770) the tool is disabled with R parameters:

1. Open the **R Parameters** page from the service menu.
2. Write **1** to the tool's parameter (150 + tool number). Example: for tool 4, **R154 = 1**.
3. To use the tool again, write **0** to the same parameter.

R parameters are retained: the value is kept when the machine is switched off and on. The values are written to the NC every time a program is started.

Notes

- **A change takes effect at the next program start.** Changing the value while a program is running does not affect the running program.
- The operations of a disabled tool are **not machined** on the part. Remember to set the parameter back to 0 / clear the box when the tool is back in use.
- The parameter has no effect for a tool that is not used in the drawing.
- Several tools can be disabled at the same time; each skips its own block.